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Smart AI Library - Intelligent Book Recommendation System (Learnova)

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Abstract

In the contemporary digital learning landscape, the challenge of delivering personalized, accessible, and time-efficient education has become increasingly critical. Modern learners face a fundamental paradox: unprecedented access to knowledge paired with diminishing time to engage with it. This project presents **Learnova** — an AI-powered Audiobook and E-Learning platform designed to bridge the gap between fragmented audiobook services and structured educational environments through intelligent automation and audio-first delivery.

The platform enables users to transform PDF books and educational materials into immersive, high-quality audio experiences via a state-of-the-art Text-to-Speech (TTS) pipeline, delivering a Spotify-inspired listening interface with persistent Mini Player support, adjustable playback speed, and chapter-level navigation. Beyond audio delivery, Learnova integrates artificial intelligence capabilities including a context-aware conversational chatbot, AI-generated chapter summarization, and a personalized recommendation engine — ensuring each user receives an adaptive and tailored learning journey.

The system is built on a modern cross-platform mobile architecture using **Flutter** and **Dart** for the frontend with Bloc state management, backed by a **PHP Laravel** RESTful API and **MySQL** database. AI features are powered by large language model APIs including OpenAI and Gemini, integrated through a LangChain-based RAG pipeline. Real-time capabilities are supported via Socket.io and Laravel Echo, while secure authentication is handled through JWT and Laravel Sanctum.

This project demonstrates the practical feasibility of unifying audio-first consumption with structured e-learning progression, establishing Learnova as a foundational model for next-generation educational platforms that prioritize accessibility, personalization, and learner autonomy.

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